



الهيئة الاتحادية للهوية
والجنسية والجمارك وأمن المنافذ
FEDERAL AUTHORITY FOR IDENTITY,
CITIZENSHIP, CUSTOMS & PORT SECURITY

ID Card Toolkit Overview

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CONFIDENTIAL

Federal Authority for Identity, Citizenship, Customs & Port Security (ICP)
United Arab Emirates

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Revision History

Version	Date	Description of Changes
1.0	28 Apr 2024	This document supersedes and replaces the "ID Card Toolkit Technical Overview" from ID Card Toolkit Release Version 3.0.1 onwards.
1.1	6 May 2024	Updated the online and offline modes.
1.2	15 Oct 2024	Updated the compatible Smartcard and Fingerprint Device List.
1.3	Mar 2026	Updated to reflect Toolkit v3.0.5 capabilities: – Added C/C++ SDK macOS platform support. – Added Cordova and Flutter SDKs. – Added Windows ARM architecture support. – Added AdES digital signature standards (CAdES/XAdES/PAdES). – Added PKCS#11 cryptographic token interface. – Added card version compatibility information.
1.4	Apr 2026	Toolkit v3.1.0.
1.5	Jun 2026	Added Dalil Information Technology AEGIS PRO 4 (AEGIS-P4-001) to the compatible device list (smartcard + biometric, USB, Contact & NFC; Windows, Linux).

Abbreviations

Abbreviation	Description
API	Application Programming Interface
CAdES	CMS Advanced Electronic Signatures
ICP	Federal Authority for Identity, Citizenship, Customs & Port Security
JWS	Java Web Start
NFC	Near Field Communication
OTG	USB On-The-Go
PAdES	PDF Advanced Electronic Signatures
PCSC	Personal Computer Smart Card
PKCS#11	Public-Key Cryptography Standards #11 (Cryptographic Token Interface)
PKI	Public Key Infrastructure
SDK	Software Development Kit
SP	Service Provider
USB	Universal Serial Bus
VG	Validation Gateway, provided by ICP
VPN	Virtual Private Network
XAdES	XML Advanced Electronic Signatures

1. Introduction

The Federal Authority for Identity, Citizenship, Customs & Port Security (ICP) created the ID Card Toolkit with Software Development Kits (SDKs) for different programming languages. The Toolkit lets Service Providers (SP) easily add Identification, Authentication (fingerprint, face, PIN, PKI), and Digital Signature features to their business applications, leveraging the capabilities of the UAE ID Card. These SDKs come with guides and examples to help developers integrate the Toolkit quickly. Even if you already use ICP's Validation Gateway (VG) legacy Java Applet and ActiveX components with your web application, switching to the Toolkit is easy. You can upgrade your application with minimal changes.

This document offers an overview of the ID Card Toolkit, designed to acquaint Service Providers (SPs) with its array of supported features, including compatible programming languages, platforms, and devices.

2. ID Card Toolkit Technical Specification

2.1. SDKs for Various Application Types

The Toolkit provides the following SDKs for different application types:

#	Toolkit SDK	Platforms	Application Types
1	Toolkit Java SDK	Windows, Linux, & macOS	Java Application
2	Toolkit .NET SDK	Windows	.NET Application
3	Toolkit JavaScript SDK & Java Web Start (JWS) Component	Windows, Linux, & macOS	Web Application
4	Toolkit Java Applet	Windows, Linux, & macOS	Web Application
5	Toolkit C/C++ SDK	Windows, Linux, & macOS	Native Application
6	Toolkit Objective-C SDK	macOS	macOS Native Application
7	Toolkit Swift SDK	macOS	macOS Native Application
8	Toolkit Android SDK	Android	Android Mobile App
9	Toolkit iOS SDK – Objective-C	iOS	iOS Mobile App
10	Toolkit iOS SDK – Swift	iOS	iOS Mobile App
11	Toolkit Cordova SDK	Android & iOS	Cordova Hybrid Mobile App
12	Toolkit Flutter SDK	Android & iOS	Flutter Mobile App

2.2. Supported Browsers

The matrix below outlines the browsers supported by the ID Card Toolkit Web/JavaScript SDK.

Platform	Chrome 70+	Firefox 78+	Edge 80+	Opera 60+	Safari 12+	IE 11
Windows 7+ (x86, x64, ARM64)	•	•	•	•	–	•
Linux 3.0+ (x64)	•	•	•	•	–	–
macOS 10.12+ (x64, ARM64)	•	•	•	•	•	–

Note: Safari is only available on macOS. Internet Explorer 11 is only supported on Windows and is deprecated by Microsoft.

2.3. Compatible Platforms & Programming Languages

The matrix below outlines the operating system support offered by each programming language specific ICP Toolkit SDK.

Platform	Obj-C	Swift	.NET	Java	C/C++	Web/JS
Windows 7+ (x86, x64, ARM64)	–	–	•	•	•	•
Linux 3.0+ (x64)	–	–	–	•	•	•
macOS 10.12+ (x64, ARM64)	•	•	–	•	•	•
Android 6.0+	–	–	–	•	–	–
iOS 9.3.5+	•	•	–	–	–	–

2.4. Supported Card Versions

The Toolkit supports the following UAE ID Card versions. Service availability may vary by card version.

Card Version	Description	Contact	Contactless
V2	Legacy ID Card	•	•
V3	Enhanced security features	•	•
V4	Current generation with ECC support	•	•
V5	Latest generation card with enhanced security	•	•

2.5. ID Card Toolkit Services

The Toolkit provides the following services. Unless explicitly specified, the service is accessible only through the contact interface for all currently active versions of the UAE ID Cards.

#	Service	Description
1	Read Public Data	Enables a Service Provider (SP) to retrieve public/open data fields from a UAE ID Card, encompassing over 90 fields including demographic, qualification, occupation, address, and passport

#	Service	Description
		information. Accessible in both offline and online validation modes with VG. The Toolkit and VG validate the data extracted from the ID Card, assuring the SP that the data remains unaltered. This service is accessible through both contact and contactless interfaces for all currently active versions of the UAE ID Cards.
2	Check Card Status	Validates and returns the status of an ID card. The status of an ID card can be active, revoked, inactive, or expired.
3	Biometric Authentication – Fingerprint	Facilitates online verification of the ID Card holder with VG by comparing live captured fingerprint data uploaded by the Toolkit against the biometric data extracted from the ID Card or the Central Database. This service is accessible through both contact and contactless interfaces for all currently active versions of the UAE ID Cards.
4	PIN / PKI Authentication	Facilitates cardholder authentication using a PIN. Authentication can occur through basic PIN verification or through a PKI challenge-response authentication method.
5	Reset or Unblock PIN	Allows an ID Card holder to reset or unblock their PIN code. PIN reset or unblocking is only permitted following successful biometric authentication of the ID Card holder.
6	Digital Signing & Verification	Empowers an ID Card holder to digitally sign documents, while also enabling Service Providers (SP) to verify signed documents. The Toolkit supports the following international Advanced Electronic Signature (AdES) standards: – PAdES (PDF Advanced Electronic Signatures) – for PDF documents. – XAdES (XML Advanced Electronic Signatures) – for XML documents. – CAdES (CMS Advanced Electronic Signatures) – for arbitrary data using CMS/PKCS#7 format. Signature levels supported include T

#	Service	Description
		(timestamp), LT (long-term), and LTA (long-term archival).
7	PKCS#11 Cryptographic Token Interface	Provides a standard PKCS#11 interface for applications that require cryptographic token access to the UAE ID Card. This enables integration with third-party applications and libraries that support the PKCS#11 standard for digital signatures, authentication, and key management operations. Supports both V2 and V4 cards with RSA and ECC key types.

3. Standard Workflow of a Toolkit Application

The table below outlines the common steps necessary for integrating the toolkit services.

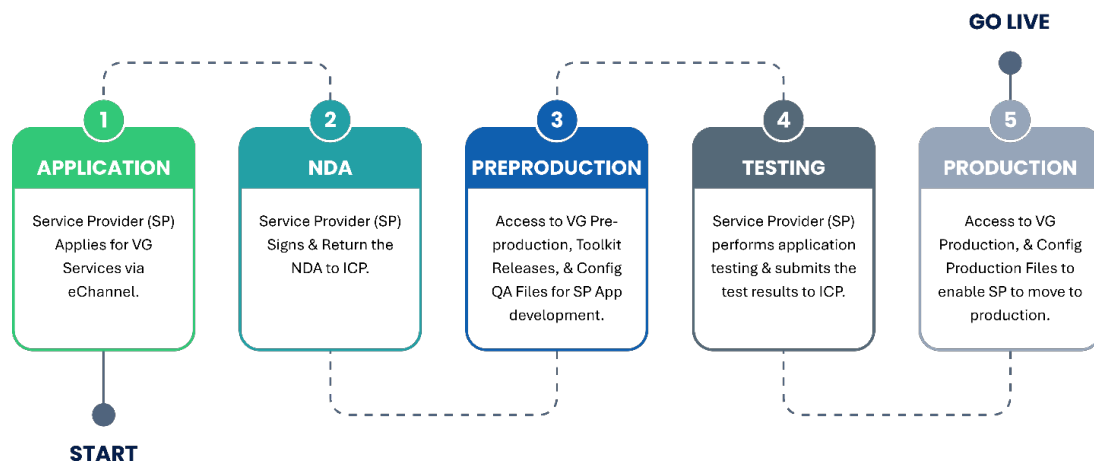
#	Step	Description
1	Initialize	The initialization process for the Toolkit varies depending on the programming language being used. In .NET, Java, JavaScript, Objective-C, and Swift, you would use the Toolkit constructor, while in C/C++, the Initialize method is used. This initialization step is crucial and should occur only once throughout the application's lifetime. During this initialization, the Toolkit must be supplied with essential application configuration parameters and toolkit config files obtained from ICP. ICP issues two distinct sets of config files to Service Providers (SP): one targeted for development and testing purposes, and another for use in production environments. The Development/QA configuration files are intended for testing the application with a test ID Card issued by ICP and conducting online validations using a pre-production VG. The Production configuration files are employed when utilizing production ID Cards and online validations using production VG.
2	ListReaders (Optional)	Lists the names of all connected smartcard readers compatible with the Toolkit. Please refer to Annexure-1 for the list of compatible smartcard readers. Note: The ListReaders, Connect, and Disconnect steps are only necessary for Toolkit services that involve communication with the ID Card.
3	Connect (Optional)	In this phase, the application endeavors to establish a connection to the ID Card in a designated smartcard reader, using one of the reader names acquired in the preceding step. Upon successful connection, the application acquires a handle to the reader, which is essential for subsequent Toolkit API functions. This connection process should be executed each time an ID Card is inserted into the card reader or when the reader is unplugged and then reconnected. Once connected, the application gains access to a variety of Toolkit services for further operations.

#	Step	Description
		Note: The process of connecting with the ID Card may vary slightly depending on the type of card interface (Contact/Contactless) being used.
4	Consume the Toolkit Services (e.g. ReadPublicData)	The Toolkit delivers service responses in a digitally signed XML format, encompassing a Request ID issued by the Application. It is incumbent upon application developers to authenticate the Toolkit Response signature and confirm that the Request ID matches the one provided to the Toolkit service. This approach ensures the authenticity of Toolkit responses, enabling applications to maintain robust security measures.
5	Disconnect (Optional)	After completing the validation process for the ID card, the application should invoke the Toolkit service to smoothly disconnect the card. Following the disconnection, the ID card can be safely removed from the reader.
6	Cleanup	In .NET, Java, JavaScript, Objective-C, and Swift, the cleanup process is automatically handled by the Toolkit destructor when the Toolkit object is destroyed. However, in C/C++, the application must explicitly call the cleanup method. Usually, this cleanup operation is conducted once during the application's lifetime, such as during shutdown. This step effectively resets the Toolkit service context and associated states, ensuring a clean and orderly termination.

4. Toolkit & VG Integration Requirements

4.1. Toolkit & VG Integration Process (Online Mode)

The diagram below illustrates the process of integrating ICP Toolkit & VG by Service Providers (SP):



4.1.1. Application Stage

During this stage, a Service Provider (SP) logs into eChannel and submits the application for VG Services. The steps involved in applying through eChannel include providing license and company registration details, specifying required Toolkit/VG Services, uploading necessary documents, and making payment for application fees.

4.1.2. NDA Stage

Service Provider (SP) signs the NDA provided by ICP and submits it to ICP.

4.1.3. Pre-production Stage

ICP provides access to the VG Pre-production, Config QA Files, and Toolkit release packages. Additionally, Service Providers (SPs) can request test cards from ICP for use in application development and testing.

Access to VG Pre-production is facilitated by ICP through VPN, with guidance provided to the Service Provider's network team for VPN setup. ICP also shares device registration credentials and login details for accessing the Toolkit Portal. This portal serves as the interface through which Service Providers (SP) can access Toolkit release packages and config files.

With the pre-production environment set up, Service Providers (SP) can build application use cases.

4.1.4. Testing Stage

The Service Provider (SP) conducts testing of the application's use cases and submits them to ICP in the prescribed format, for review and approval.

4.1.5. Production Stage

Upon approval of the test results, ICP grants access to VG Production and Config Production Files. Access to VG Production is established through an MPLS connection, which requires coordination between the Service Provider (SP) and Etisalat for the related process.

Once the VG Production connectivity is established, the application can be deployed to production along with the Production Config.

4.2. Toolkit Integration Process (Offline Mode)

To utilize the ID Card Toolkit in offline mode, a Service Provider (SP) must download the Toolkit along with the offline configuration files and integrate them with the business applications.

In offline mode, the Toolkit solely permits the reading of open public data from the card. Services such as checking card status and biometric authentication are exclusively available in online mode.

5. Annexure-1: Compatible Smartcard & Fingerprint Devices

The table below lists the smartcard and fingerprint readers compatible with the Toolkit. Before purchasing a device, please review the following notes carefully.

Notes:

1. The information presented in the table is based on the data accessible to ICP at the document's release. It's important to note that a device manufacturer may release updates for a device while retaining the same name. However, these updates may not be compatible with the Toolkit.
2. Prior to purchasing devices in bulk, it is advisable to take two key steps. Firstly, confirm compatibility with the specified device SDK versions by reaching out to the manufacturer or reseller. Secondly, conduct tests with a sample of the specific device using the Toolkit to verify that any hardware or firmware updates have not compromised compatibility.

#	Vendor	Model	Features / Capability	Connect	Interface	Platforms
1	Morpho	MSO 1350 (v2, v3)	Smartcard Reader (CCID USB Plugin), Biometric Reader (MorphoSmart SDK 6.14.5.0)	USB	Contact	Windows
2	Morpho	MSO 1350 (v2, v3)	Biometric Reader (MorphoSmart SDK 6.14.5.0), Smartcard Reader (Not Supported)	USB	Contact	Linux
3	Feitian	bR301	Smartcard Reader (Feitian SDK v1.31.5)	Bluetooth	Contact	Windows, Android & iOS
4	Feitian	bR301	Smartcard Reader (Feitian SDK v1.31.5)	USB	Contact	Windows, Linux & macOS
5	Feitian	iR301-u	Smartcard Reader (Feitian SDK v1.31.5)	Lightning	Contact	iOS
6	Feitian	iR301-uc	Jacket type Smartcard reader	Lightning	Contact	iOS

#	Vendor	Model	Features / Capability	Connect	Interface	Platforms
			for iPads (Feitian SDK v1.31.5)			
7	ACS	ACR38u	Smartcard Reader (ACS SDK 1.1.4)	OTG	Contact	Android
8	ACS	ACR 122u	Smartcard Reader (CCID USB Interface)	USB	NFC	Windows
9	Smart Phones	Android NFC phones	Smartcard Reader (Android NFC interface)	N/A	NFC	Android
10	Any	PCSC CCID compatible	Smartcard Reader (CCID USB Interface)	USB	Contact	Windows, Linux & macOS
11	TazTag	TazPad ID (v1 & v2)	Tablet + Inbuilt: Smartcard Reader (ACS SDK 1.1.4), Biometric Reader (MorphoSmart SDK 6.14.5.0 / 6.22.0.0)	N/A	Contact	Android
12	Safran Morpho	Morpho Tablet 2S	Tablet + Inbuilt: Smartcard Reader (ACS SDK 1.1.4), Biometric Reader (MorphoSmart SDK 6.14.5.0 / 6.22.0.0)	N/A	Contact	Android
13	Gen2wave	RP1600	Handheld + Inbuilt: Smartcard Reader (RP1600 SDK)	N/A	Contact	Android
14	Gen2wave	EID 10 Plus	Handheld + Inbuilt: Smartcard Reader, Biometric Reader	N/A	Contact	Android

#	Vendor	Model	Features / Capability	Connect	Interface	Platforms
			(Combi Plugin for EID 10 Plus SDK)			
15	Gen2wave	RP70A Bio	Tablet + Inbuilt: Smartcard Reader, Biometric Reader (Combi Plugin for RP70A Bio)	N/A	Contact	Android
16	Smart Phones	iPhone (NFC)	Smartcard Reader (iOS NFC interface). iPhone 7+ with iOS 13+	N/A	NFC	iOS
17	Identos GmbH	Tactivo Mini (Android)	Smartcard Reader, Biometric Reader (Combi Plugin SDK v2.0.2)	USB	Contact	Android
18	Grabba	S Series	Jacket and USB type: Smartcard Reader, Biometric Reader (Plugin SDK V1.194)	USB	Contact	Android
19	Telpo	TPS360	Handheld + Inbuilt: Smartcard Reader, Biometric Reader (Combi Plugin for TPS360 SDK)	N/A	Contact	Android
20	Ekemp	Digit9 IAM M8 Artisecure Suprema	Tablet + Inbuilt: Smartcard Reader (Artisecure SDK v1.0.0), Biometric	N/A	Contact	Android

#	Vendor	Model	Features / Capability	Connect	Interface	Platforms
			Reader (Suprema BM-Slim2 SDK v20)			
21	DUNN BioPOS	BioPOS-100	Handheld + Inbuilt: Smartcard Reader, Biometric Reader (Combi Plugin SDK v1.1.2)	N/A	Contact	Android
22	DUNN BioPOS	BioPOS-100	Handheld + Inbuilt: Smartcard Reader, Biometric Reader (dsap_biopos_1.1.2)	USB	Contact	Windows
23	Secugen	Hamster Pro Duo SC PIV	Smartcard Reader (CCID USB Interface), Biometric Reader (SG_DEV_FDU03/FDU05)	USB	Contact	Windows
24	Dermalog	ZF1 Scanner	Biometric Reader (Dermalog ZF1 Scanner Plugin)	USB	Contact	Windows
25	Idemia	IDScreen Biometric Tablet	Tablet + Inbuilt: Combi Plugin for Idemia Smartcard Reader and CBM E3 Fingerprint Sensor. ID Screen SDK 1.2	N/A	Contact	Android
26	Identos GmbH	Tactivo Mini (iOS)	Smartcard Reader (SDK v2.23)	USB	Contact	iOS
27	Feitian/ Chengxin	Paysky Smartbox v7 POS	Handheld + Inbuilt: Combi Plugin for	USB	Contact	Android

#	Vendor	Model	Features / Capability	Connect	Interface	Platforms
			Paysky Smartcard Reader and Feitian Fingerprint Sensor. Paysky SDK v1.0.0, Feitian FP SDK v1.0.9			
28	GripID	GripID V6	Handheld + Inbuilt: Smartcard Reader (ACS SDK 1.1.5), Biometric Reader (Suprema BM-Slim2 SDK v20)	N/A	Contact	Android
29	GripID	GRIPIDFAP20	Handheld + Inbuilt: Smartcard Reader (ACS SDK 1.1.5), Biometric Reader (Suprema BM-Slim2 SDK v20)	N/A	Contact	Android
30	Tech5 (T5-AirSnap)	Android Phone, iPhone	Contactless Camera Biometric (Fingerprint Plugin support)	(Contactless Capture)	N/A	Android, iOS
31	GripID	GRIPIDFAP30	Handheld + Inbuilt: Smartcard Reader (Alcor Micro), Biometric Reader (IBScanUltimate_Android_4.3.0)	N/A	Contact	Android
32	Dalil Information Technology	AEGIS PRO 4 (AEGIS-P4-001)	Smartcard Reader, Biometric Reader (Firmware RT-FW-AEGIS-	USB	Contact & NFC	Windows, Linux

#	Vendor	Model	Features / Capability	Connect	Interface	Platforms
			P4-v0.9.5)			